CertaRig: Reviewed Commissioning and Verifiable Rig Evidence

CAPSTONE SOFTWARE CHECKPOINT — REVIEW DRAFT

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Approved baseline: CertaRig Phase 3 signed report, 19 September 2026. Current capstone supervisor assignment, academic year, final deadline and submission process require student confirmation.

Evidence boundary: local software, deterministic fixtures and simulated plants. The new fluid loop has not been purchased, assembled or physically qualified. Live-provider comparison is deferred.

Declaration

Student review and signature pending. No signature or originality attestation is made by this draft. Review all institution-required declarations before submission.

Acknowledgements

The approved Phase 3 report records prior supervisor guidance. Further acknowledgements should be completed by the student from actual contributions; no new meeting or feedback is inferred.

Abstract

CertaRig is a local rig workflow that connects configuration review, readiness checks, operator approval, deterministic execution and portable evidence. This capstone extends the approved Phase 3 dry-bench foundation toward a supported, already-mapped rig rather than attempting unrestricted control of unfamiliar equipment. The implemented software introduces resumable commissioning interviews with source references and explicit blockers for incomplete mappings, unknown units, missing calibration evidence and unsupported procedures. Only an operator can apply the proposed configuration or grant execution authority.

A distinct first-order thermal simulator models heat capacity, ambient heat loss and an external heating disturbance using deterministic time steps and seeded measurement noise. Semantic configuration comparisons produce conservative revalidation requirements. Simulator rehearsal receipts are bound to the procedure revision, normalized test profile and model implementation revision. A broker-backed MQTT adapter provides observation only and rejects malformed, retained, stale and disconnected readings.

Run bundles preserve raw samples, configuration history, interview provenance, calibration references, approvals, events and reproducible checks. Offline verification rejects missing or corrupted evidence and recalculates supported measurements. Mixed-sensing and electrical-feedback adapter foundations and staged physical-review gates are implemented and tested with software fixtures. These tests do not establish real wiring, calibration or actuator performance.

The proposed physical increment is a protected, room-temperature water loop with independently measured feedback. Procurement remains gated by a corrected pressure-sensor specification, matched component ratings and delivered costs within a fifteen-thousand-rupee ceiling. Physical commissioning, attended endurance measurements and live-model comparison remain outstanding. This report records the software checkpoint and the evidence required to complete the final capstone.

Keywords: commissioning; deterministic control; thermal simulation; evidence integrity; operator approval; measured feedback.

Contents and Preliminary Lists

1. Introduction 4

2. Implementation Details 5

3. Testing, Validation and Results 8

4. Execution and Deployment Details 11

5. Project Execution Evidence 12

6. Conclusion and Future Work 13

References 14

Appendix A: User Manual 15

Appendix B: Installation and Demo Guide 16

Page references apply to this review PDF. Refresh them after student revisions.

List of Tables

Table 1: Current validation. Table 2: Assessment evidence. Table 3: Progress and supervisor record.

List of Figures

No numbered figures are included in this draft; the accompanying editable presentation and browser screenshots show the architecture and operator workflow.

Abbreviations

ADC: analog-to-digital converter; API: application programming interface; CSV: comma-separated values; HIL: hardware in the loop; MQTT: Message Queuing Telemetry Transport; RC: resistance/capacitance analogue; SDK: software development kit; UTC: Coordinated Universal Time.

CHAPTER 1: INTRODUCTION

1.1 Background and problem

The approved Phase 3 report establishes a dry electronic bench, local safety/control boundaries and evidence-backed procedures. Its seven future missions guide the capstone: conversational commissioning, a distinct second plant, complete evidence, measured feedback, model qualification, broader integrations and calibrated protected actuation. Approval of that report is a verified baseline; it is not proof that later hardware work has occurred.

A configuration file alone cannot establish readiness. An operator must understand the signal identity, units, scaling, calibration and operating limits, then review what is proposed. A result that refers to a raw recording outside its export folder is not independently reviewable. Outstanding execution authority must not survive a restart or an unreviewed material change.

1.2 Objectives and scope

The target workflow is: describe a supported rig, review its mapping, verify readiness, rehearse its procedure, approve a run, measure the response and export verifiable evidence. One Edge instance has one active rig. The initial supported profiles are pressure/flow simulation and a distinct thermal simulator; onboarding changes configuration rather than application source.

The user selected a subsequent low-energy water-loop extension with electrical feedback within ₹10,000–15,000. The software checkpoint is 20 November 2026. New hardware, deployment to the frozen Phase 3 installation, unattended physical operation, paid model calls, production hosting and arbitrary-machine discovery are outside this implementation turn. The institution’s final deadline remains unconfirmed.

1.3 Expected deliverables

Deliverables comprise reviewable source, contracts and reproducible tests; the two simulator journeys; observation-only broker integration; a portable evidence bundle; a six-chapter report and ten-slide deck; genuine progress records; and a gated physical design/procurement review. Final completion additionally needs real calibration, cutoff and integrated-loop evidence, actual supervisor review and the student’s narrated demonstration.

CHAPTER 2: IMPLEMENTATION DETAILS

2.1 Architecture and authority

Studio and the provider-neutral agent are clients of the same Edge API. The Edge enforces principal identity, contract revision, reviewed configuration and exact-argument approval. Local procedures and guardrails execute and record independently of model availability. The agent may collect answers and propose changes; it cannot apply mappings, accept physical commissioning or grant authority.

Existing atomic-file approval storage and complete run exports are retained from baseline 2f25d66. Grants are bound to the requesting principal, exact procedure/arguments, configuration and procedure revision. A grant is consumed once. Restart expires outstanding authority while retaining audit history. Configuration application invalidates old grants and commissioning review.

2.2 Resumable commissioning interview

Interview records live separately from active configuration and are saved atomically with optimistic revision checks. Answers retain source references, unresolved text and prior revisions. The interview covers equipment, interfaces, signal identity, units, scaling, calibration references, limits, safe state and supported procedures through a structured rig document and bounded labelled answers. Unknown values are not guessed.

Create/read/update/propose operations are exposed under /v1/commissioning/interviews. A candidate profile must explicitly require commissioning. Required calibration IDs need supporting references; unknown source identities, incompatible procedure units, missing limits, unsupported procedures and unresolved lines prevent proposal. Proposing does not change the active rig. The operator reviews the proposed diff and applies the exact interview revision through /v1/rig/apply.

The applied interview is snapshotted against its configuration hash. Both supported simulator templates are explicitly labelled synthetic; their calibration references do not represent laboratory calibration. In the fixture agent, labelled lines and JSON are interpreted deterministically. Provider-neutral tools permit future qualified language interpretation without extending authority.

2.3 Distinct thermal model

The opt-in thermal_rc plant uses C dT/dt = Q − k(T − Ta). With fixed step dt and k greater than zero, the exact discrete update approaches Ta + Q/k exponentially at rate k/C. The recorded default uses C=100 J/°C, k=2 W/°C, ambient 25 °C, initial 45 °C and external heat input 40 W. Seeded measurement noise is independent of the deterministic state evolution. No physical heater is added.

Legacy profiles retain the generic simulator. Cooling, equilibrium, external overtemperature, dropout and stale-source scenarios exercise different expected behaviors. Model parameters, seed, sampling interval and implementation revision thermal-rc-v1 identify the simulation used in a rehearsal.

2.4 Semantic changes and rehearsal identity

Configuration comparison aligns channels by identity and examines source mapping, units, scaling, calibration and limits. A changed measurement claim invalidates its associated qualification and requests identity, calibration, range/quality and procedure-repeat checks. Rig-level or unclassified changes require full qualification. Previous results and configurations remain available as history; this rule set does not claim the globally smallest retest set.

A twin receipt binds procedure hash to a normalized test-profile hash and simulator model revision. The profile preserves mappings, calibration, limits, control semantics, wiring revision and model parameters while excluding deployment mode. A changed profile or implementation revision cannot reuse the old receipt. Receipts also expire. A passing first simulator run is a rehearsal; a later explicitly approved run records that prior rehearsal in its evidence.

2.5 Observation-only MQTT

The optional Paho MQTT backend makes an actual broker connection. Exact topics map to reviewed channels. Messages include numeric value, matching unit and timezone-aware source timestamp. The adapter exposes connection state and rejects missing, malformed, nonfinite, retained, stale, future and disconnected observations. Local receipt time never substitutes for source time.

Network mode is observation only and contains no equipment-control publish path. Local force-safe does not publish a remote command and does not claim to know remote output

state. The existing in-process TagBus remains a legacy test fixture. Authentication can use environment references and TLS; the tested anonymous loopback Mosquitto instance is not a production deployment.

2.6 Mixed sensing and measured feedback foundations

The Pi adapter supports calibrated ADC acquisition, pulse-flow counting, explicit 1-Wire temperature identity with CRC checks, and INA219 shunt-voltage conversion. Raw measurements, coefficients, acquisition timestamps and quality remain separate. Zero pulses can be a valid fresh zero-flow observation; absence of a successful acquisition is a quality fault. Source-specific freshness accommodates different acquisition cadences.

Background per-source readers keep blocking device reads out of the deterministic snapshot path. Cached samples age out; the software cannot terminate a wedged operating-system I/O transaction. Independent actuator cutoff remains required. Required electrical feedback is recorded separately from command. Missing or mismatched feedback forces safe and retains the trip latch. Sampled transition intervals express acquisition uncertainty; current proves electrical response, not mechanical valve position.

2.7 Staged physical review and portable evidence

A separate operator API records observation, attended trial and accepted stages. Records bind configuration, adapter, wiring revision, calibration, reviewer, supported procedures and qualification evidence. Physical check files require raw artifacts with verified local hashes and exact configuration identity. Trial authority expires after thirty minutes or restart. Accepted procedures require passing physical trial records. Direct capstone Pi reset/permit commands are blocked; output procedures must declare their need for output authority.

The API checks consistency of the operator attestation and supplied files. It does not certify that a laboratory claim is true. No real physical acceptance is issued here. Run bundles embed relevant physical evidence for portability when such evidence exists. Interview, configuration history, calibration, revalidation, rehearsal, approval, raw CSV, results, events and checksums accompany the current simulator bundle.

CHAPTER 3: TESTING, VALIDATION AND RESULTS

3.1 Method and environment

Validation uses targeted contracts, existing unit and agent tests, property-based invariants, deterministic simulator scenarios, browser journeys, a real local broker, a separate accelerated soak and an isolated clean editable installation. The existing 90 percent coverage threshold is retained. Hardware-in-the-loop and live-provider tests are excluded from the software pass claim. Test fixtures labelled physical exercise record validation only; they are not real physical evidence.

The environment uses Python 3.14.3 and the project's existing Ruff, mypy, pytest, Hypothesis, coverage and Playwright tooling. The broker is eclipse-mosquitto:2.0 on loopback port 18883. Reproduction commands and raw receipts are included in the handover. The accelerated hour is simulated time; it is not sixty minutes of attended hardware acquisition.

3.2 Acceptance design

Commissioning contracts test resumable drafts, revision conflicts, source/calibration/unit/identity blockers, unsupported procedures, operator-only application and configuration-bound provenance for both profiles. The agent tests exercise deterministic interview routing, refusal, continuation and proposal without authority. Existing approval contracts cover denial, expiry, replay, wrong principal, mismatched configuration/arguments/revision and restart history.

Thermal tests compare the numerical state against the analytical RC solution and exercise cooling, steady state, heat trip, dropout and stale data. Revalidation tests change channel mapping/calibration and model identity. MQTT callback and real-broker tests demonstrate valid receipt, staleness, disconnection, bad messages and absence of command publication.

Feedback and physical-review tests use injected software adapters: zero-flow acquisition, invalid sources, CRC failure, nonfinite reads, missing/mismatched feedback, staged check-file identity, tamper, trial expiry and required output declarations. These tests establish software behavior only. Browser acceptance additionally reloads an incomplete interview, restores it and applies both supported profiles through the UI.

3.3 Current results

Table 1: Current validation

Check	Observed result	Evidence boundary
selected software tests	283 passed; 1 opt-in broker test skipped in default suite	Local software / fixtures
lint and typing	pass; 80 Python source files checked	Local software / fixtures
coverage	90.40% (unchanged gate 90%)	Local software / fixtures
browser	8 passed	Local software / fixtures
local broker	2 passed with CERTARIG_MQTT_TEST_PORT=18883	Local software / fixtures
accelerated simulator soak	1 passed; not attended physical uptime	Local software / fixtures
clean editable install	pass; isolated new environment; CLI help works	Local software / fixtures
portable interview/rehearsal/run bundle	pass; offline verified	Local software / fixtures

The current automated demo run is run_20261109T092836_b80741. Its previous rehearsal is run_20261109T092835_2dddf4. The exported raw recording has 77 rows: 75 sample rows and 2 command rows. Recorded jacket temperature spans 44.87777 to 45.12445 °C. Offline analysis found 0 unsafe command rows and 0 electrical feedback observations. The absence of physical feedback is explicit, not interpreted as a successful measurement.

The demo confirms an incomplete interview cannot propose an executable profile, unreviewed execution is refused and a denied approval cannot start a run. A fresh automated fixture grant completes the simulated run. The export verifies offline. Automated grants validate API behavior; the student must perform actual operator review in the final demonstration.

3.4 Independent verification and limitations

The verifier checks checksums, required manifest entries, recording inclusion/count, approval provenance and configuration/interview consistency. It recomputes row counts, extrema and supported peak, event and step checks from raw samples and recorded events. Removing artifacts or changing a recorded summary fails verification. Tampered and malformed bundles preserve structured failure reasons.

Recorded software-event checks are not an independent physical observation. Latency checks unsupported by the raw recording are explicitly marked not independently reproduced. Checksums provide bundle integrity, not external authorship. The final physical report must add reference uncertainty, calibration repeatability, electrical response intervals, flow response and the attended sixty-minute dataset. No mechanical relay timing or valve-position accuracy is inferred from this simulator result.

3.5 Findings and recovery

Integration checks exposed stale UI contract handling, provenance-update consistency, raw summary compatibility and malformed-evidence reporting cases. The implementation preserves old summaries by comparing their recorded fields to current calculations; new bundles add explicit check reproduction. Unknown or unclassified changes take the conservative full-requalification path. Historical successful runs remain history and cannot replace the active contract.

A development browser navigation can close an HTTP socket before a poll response; the server logs a broken pipe during teardown while the browser suite passes. No result is interpreted as a physical success. The raw validation logs accompany this report so failed intermediate checkpoints are not confused with the final release results.

CHAPTER 4: EXECUTION AND DEPLOYMENT DETAILS

4.1 Installation and operating sequence

The implementation remains on an isolated worktree based at 2f25d66, preserving the research checkout and frozen Phase 3 installation. A clean editable install with development and MQTT extras is validated separately. The local simulator binds loopback and uses private runtime/evidence folders. It does not deploy to the course Pi or create cloud hosting.

The operator completes the source-backed interview, reviews/applies the proposal, reviews readiness, approves rehearsal and later grants a fresh execution. A separate process verifies the export. The five-minute script includes denial and recovery. Both profiles onboard without application-source edits.

4.2 Hardware design and procurement gates

The proposed loop combines an open reservoir, flooded low-head pump, pressure reference, pulse flow, temperature/current sensing, a zero-differential NC valve, permanent bypass, pressure protection and containment. A separately fused actuator supply and independent cutoff remove actuator power while Pi recording continues.

Procurement is on hold. The PR19C-PA-050K sheet conflicts between a 0–50 kPa label and a table ending at 20 kPa. A FESCOLO 2W025-08 direct-acting valve and low-head TOPSFLO pump are candidates; exact variants, delivered quotes, reference/protection and current wiring still need review. HARDWARE_REVIEW.md records interfaces and source evidence.

No physical pressure is frozen. Select operating limit, software trip, independent protection, weakest component working rating and uncertainty together. Simulated 4.2 bar is not a physical target. Keep ADC inputs within supply rails. Review the complete delivered basket within ₹15,000 before ordering; ₹13,000 is an allocation, not a quote.

4.3 Physical deployment sequence

Use a separate microSD and reviewed circuit. Sequence: dry feedback, passive leak checks, read-only sensing, calibration, protected trial, integrated procedures. Verify physical E-stop and no automatic restart. Short cycles precede the attended sixty-minute session. Material changes require renewed acceptance.

CHAPTER 5: PROJECT EXECUTION EVIDENCE

5.1 Source and assessment traceability

Actual commits, patch and mission/rubric mapping accompany the handover. Source: https://github.com/shadybrook/certarig-platform . Changes remain local; no remote publication or university submission is claimed.

Table 2: Assessment evidence

Category / marks	Available evidence	Still required
Continuous evaluation / 20	Dated work log; signed Phase 3 baseline	Actual weekly supervisor feedback
Implementation / 30	Source, contracts, simulator, broker, UI	Physical measured integration
Testing / 15	Current automated logs and bundle	Calibration, endurance and uncertainty
Documentation / 20	Six chapters, guides, references	Student review and final PDF submission
Presentation / 15	Ten-slide deck and timed script	Actual rehearsal and narrated demo link

5.2 Genuine progress and supervisor interaction

Table 3: Progress and supervisor record

Date	Verified record	Supervisor remark
19 September	Signed Phase 3 baseline	Historical approval
9 November	Software, tests and sourcing review	
10–20 November	Planned review and rehearsal	

New supervisor remarks remain blank. The update is unsent. Confirm the current supervisor, final deadline and submission process. Historical approval is not a new capstone meeting.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Achieved software outcome

The checkpoint connects resumable source-backed commissioning, operator-reviewed configuration, readiness, exact approval, deterministic simulated execution and portable evidence. A distinct thermal plant and observation-only broker extend the prior dry-bench foundation. Rule-based revalidation prevents a changed measurement claim from silently retaining prior qualification. Mixed-source and physical-review foundations are reviewable code with software tests.

The final capstone is not physically complete. The water-loop basket, current wiring design, actual calibration, independent power-cutoff validation and attended endurance evidence remain outstanding. Live-provider comparison is a disclosed deferred mission, not replaced by claims about deterministic fixtures.

6.2 Exact next actions

Obtain one separate microSD. Confirm the institution’s final deadline/reviewer/submission requirements. Supply current bench photographs for the wiring/design review. Resolve the pressure-transmitter specification and exact pump/valve/protection/reference quotes, then review the complete delivered basket before any order. Schedule attended dry feedback, sensing/calibration and protected wet-loop sessions after design acceptance. Retain real supervisor feedback and rehearse the supplied fifteen-minute script.

6.3 Further development

After physical commissioning and the software checkpoint, qualify real Python/MQTT equipment sources, run actual candidate-provider evaluations on the prepared cases, and separately accept any authenticated remote pilot. No unattended physical operation follows automatically from a simulator pass. Improve reference uncertainty and source-health detection using measured evidence rather than increasing simulated thresholds.

REFERENCES

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APPENDIX A: USER MANUAL

Open the local Studio and sign in with the simulator's local demo credentials. Select Onboard, create a guided interview and supply equipment, safe state, procedures, the structured map and calibration sources. Save the draft and deliberately propose it incomplete to see explicit blockers. Reload and resume; no executable configuration has changed.

Load either simulator template and review every signal/source/unit/limit. Keep synthetic references labelled. Save and propose, inspect the diff/revalidation checklist, then apply as operator. Review the simulated configuration for the chosen procedure. Request and deny a run to show refusal. Request a fresh rehearsal, inspect the exact procedure/arguments and grant once. Repeat with fresh execution approval after rehearsal. A changed mapping requires renewed review and approval.

Inspect the run's raw recording and displayed result in Evidence, export this run and verify the downloaded ZIP offline. Never repair a failed evidence check by editing only its checksum list; investigate missing/corrupt sources. Physical controls remain disabled pending the separate physical design and acceptance gates. Force-safe remains available locally; releasing E-stop never authorizes automatic restart.

APPENDIX B: INSTALLATION AND DEMO GUIDE

Use the installation commands in EXTENSION.md from this branch. A new private runtime folder and clean copied simulator configuration keep tests separate from historical evidence. Install the MQTT extra for broker observation; the local Mosquitto recipe binds loopback only. The isolated clean-install receipt records the environment actually checked.

Reproduce the scripted bundle with tools/run_capstone_demo.py using a fresh --work directory, --out directory and the explicit --automated-approval-fixture flag. Verify with: .venv/bin/python -m certarig evidence verify <bundle.zip>. This automated fixture is not a real human approval or spoken rehearsal.

The ten-slide deck follows the supplied university template. Allocate 0:00–5:45 to introduction/objectives/architecture/tools, 5:45–10:45 to live success/refusal/export, and 10:45–15:00 to results/challenges/remaining work. The live segment shows an incomplete interview, corrected proposal, operator review, denial, fresh approval, completed run, export verification and damaged-copy refusal. Use the prepared bundle/screenshots only as a disclosed fallback.

Current local demo link: http://127.0.0.1:18090 . This link works only while the local process runs. Public narrated video link: pending student recording and approved hosting. Final institution submission link and deadline: pending confirmation. No hosted demo or narrated recording is fabricated.